Open Source Software Capstone Project

<u>The Open Source Audit</u>

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Course: Open Source Software

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1. Introduction

Open source software plays an important role in today's digital world.Many systems and applications depend on open source tools for development and execution.

In this project,I selected VLC Media Player as my chosen software. It is a widely used open-source multimedia player. It supports various formats and platforms.VLC is known for its simplicity,flexibility and powerful features.

This project focuses on understanding the origin,license,ethical aspects and technical working of VLC Media Player. Along with that,some Linux based commands and shell scripting tasks are also performed.

Part A1: Origin of VLC Media Player

VLC Media Player is a widely used open-source multimedia player. It supports various formats and platforms.VLC Media Player is a very popular software used for playing videos and audio files.

But it was not created randomly,there was a reason behind it. It was first developed by students from a university in Paris.They wanted to stream videos inside their campus network, but there was no proper software available at that time.Most of the tools were either paid or limited.

So they decided to build their own media player which can handle different formats and also support streaming.This is how VLC started. It was a small project at beginning,but later it became very powerful.

One important thing is that VLC is open source.It means anyone can see the code, modify it and improve it.This helped the software to grow faster because many developers from different places contributed.

Today VLC is used by millions of users.It supports almost all video formats and works on Windows,Linux and Mac. Many people prefer VLC because it is free,simple and powerful.

In my opinion,VLC shows how open source software can solve real problems. Instead of depending on companies,people can create and share their own solutions.

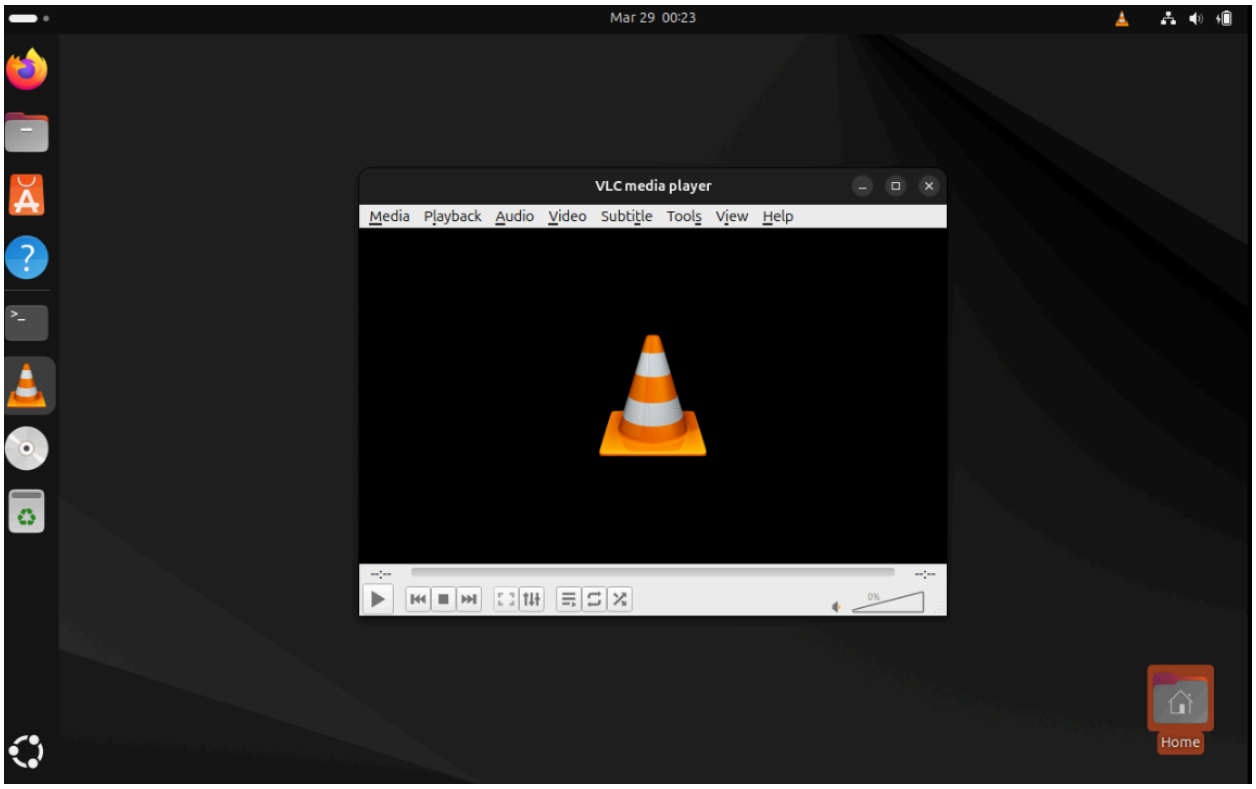
VLC also became popular because it does not require additional codecs.Users can directly play almost all types of media files without installing anything extra.This makes it very convenient for beginners.

VLC also became popular because it does not require any external codecs.Users can directly play almost all types of media files without installing anything extra. This makes it very easy to use even for beginners.

Another reason for its success is its lightweight nature.VLC does not consume too much system resources,so it can run even in low end systems without any issue. This makes it different from many heavy media players.

Over time,VLC has added many features like subtitle synchronization,video streaming and screen recording. Even though it started as a student project,it has now become a professional level software used worldwide.

Figure 1: VLC Media Player running on Ubuntu system



Part A2: License Analysis of VLC Media Player

VLC Media Player uses open source licenses like GPL and LGPL. These licenses give users freedom to use,study and modify the software.

There are four main freedoms in free software.The first freedom is to run the program for any purpose. The second is to study how it works.The third is to modify it and the fourth is to share it with others.

VLC follows these freedoms,so it can be considered as a true open source software.

If a company takes VLC and modifies it,they may need to share the changes also if they are using GPL.This is different from MIT license, where companies can modify and keep it private.

There is also a difference between free as in free beer and free as in freedom. Free beer means no cost,but freedom means control over the software.VLC gives both.If I create my own software,I would choose GPL license because it ensures that my work remains open for others.

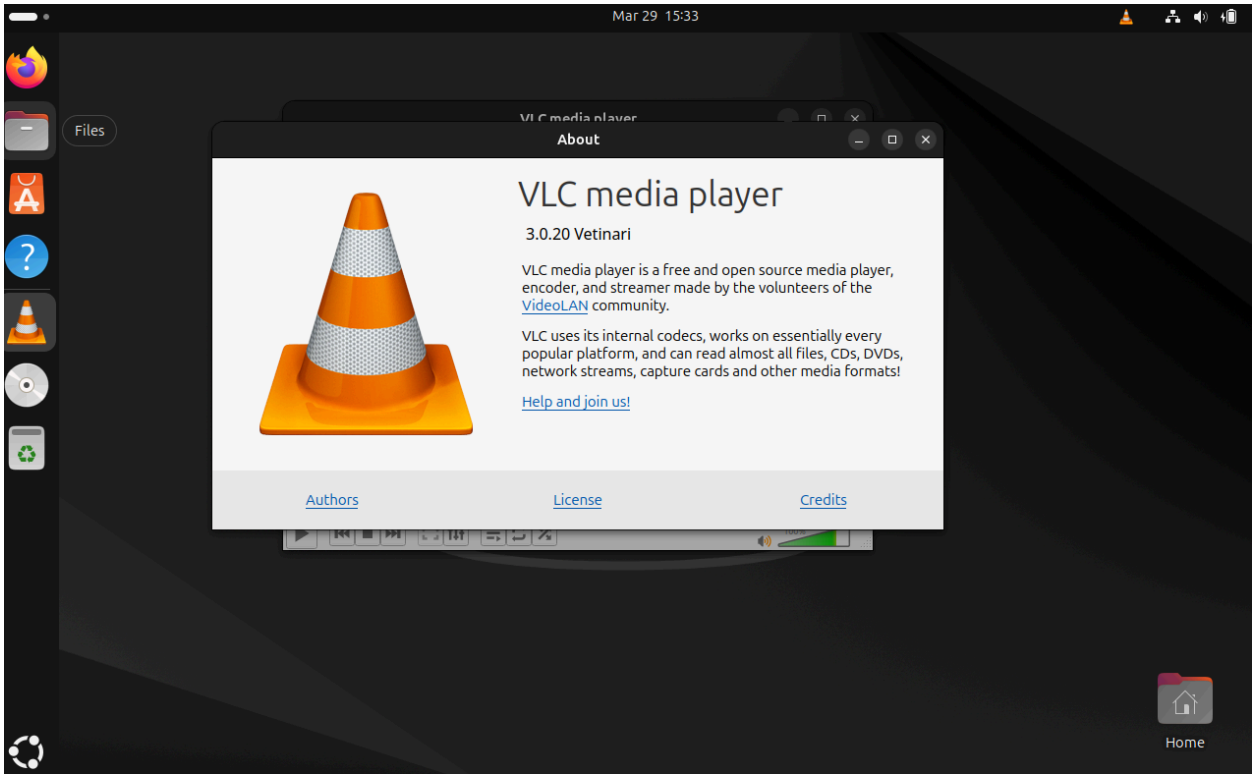
Another advantage of GPL license is that it protects user freedom.It ensures that the software remains open and prevents misuse by private companies.

Another important point about open source licenses is transparency.Users can trust the software because they can see how it is built. This reduces the risk of hidden features or malicious code.

GPL license also ensures that future versions of the software remain open.If someone improves VLC and distributes it,they are expected to share the source code as well. This helps the community grow continuously.

In comparison,some proprietary software do not give users any control over the code.Users can only use it but cannot understand or modify it.This is where open source licenses become very important.

Figure 2: VLC version and license details



Part A3: Ethics of Open Source Software

Open source software is not only about code.It is also about sharing knowledge and improving together,in many cases this becomes very useful.

Not all software needs to be open source.Some companies may want to keep their code private for business reasons,but important tools should be open so everyone can use and improve them.I feel this balance is necessary.

There is also a question whether companies should earn money using open source software.I think it is acceptable if they also contribute back to the community,otherwise it may feel unfair.

The idea of standing on the shoulders of giants means we are building on previous work.Open source makes this easier because we can use and improve existing tools,and this saves a lot of time also.

So overall,open source supports innovation rather than stopping it.Open source also helps students and beginners to learn real world coding.By studying open source projects,they can improve their knowledge and skills,and this is something I personally feel important.

Open source also encourages collaboration.People from different countries can work together on the same project and improve it.This creates a strong global community,but coordination is also needed sometimes.

It also reduces dependency on big companies.Users are not locked into one system and they have the freedom to choose and modify tools according to their needs,which is very helpful.

In my view,open source is not just a technical concept,it is also a mindset.It promotes sharing,learning and innovation among people,and this is one reason why it is growing continuously.

Part B: Linux Footprint of VLC Media Player

1. Installation of VLC on Ubuntu

VLC Media Player was installed on Ubuntu operating system using terminal commands.The installation process was simple and completed successfully,so there were no major issues.

The installation process did not require any complicated steps.The package manager handled all dependencies automatically,which made the process easier.

After installation,VLC was opened and tested using a sample video. This confirmed that the software is working properly without any issues,and playback was smooth.

The system was first updated using package manager.After that,VLC was installed using apt command,this is the usual method followed in most cases.

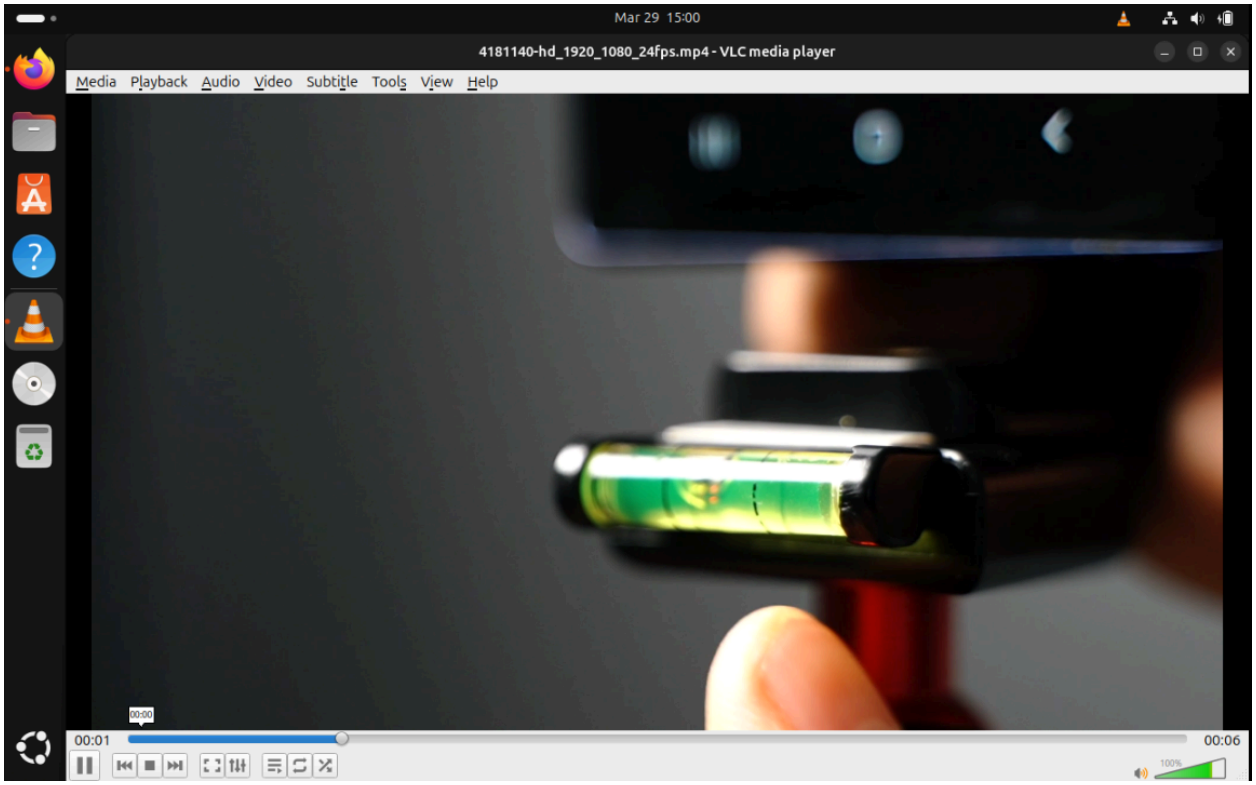
Commands used:

```
sudo apt update
```

```
sudo apt install vlc
```

The installation completed without any errors and VLC was successfully launched from the applications menu.

Figure 3: VLC Media Player playing a video on Ubuntu system



2. Basic Linux Commands Used

Basic Linux commands were used to understand file and directory management.These commands are very important especially for beginners who are trying to understand how Linux works.

They help in basic file handling and navigation inside the system,and in most cases these are the first commands users learn.

By using these commands,it becomes easier to manage directories and files without using graphical interface.This improves efficiency and gives more control over the system,which is useful.

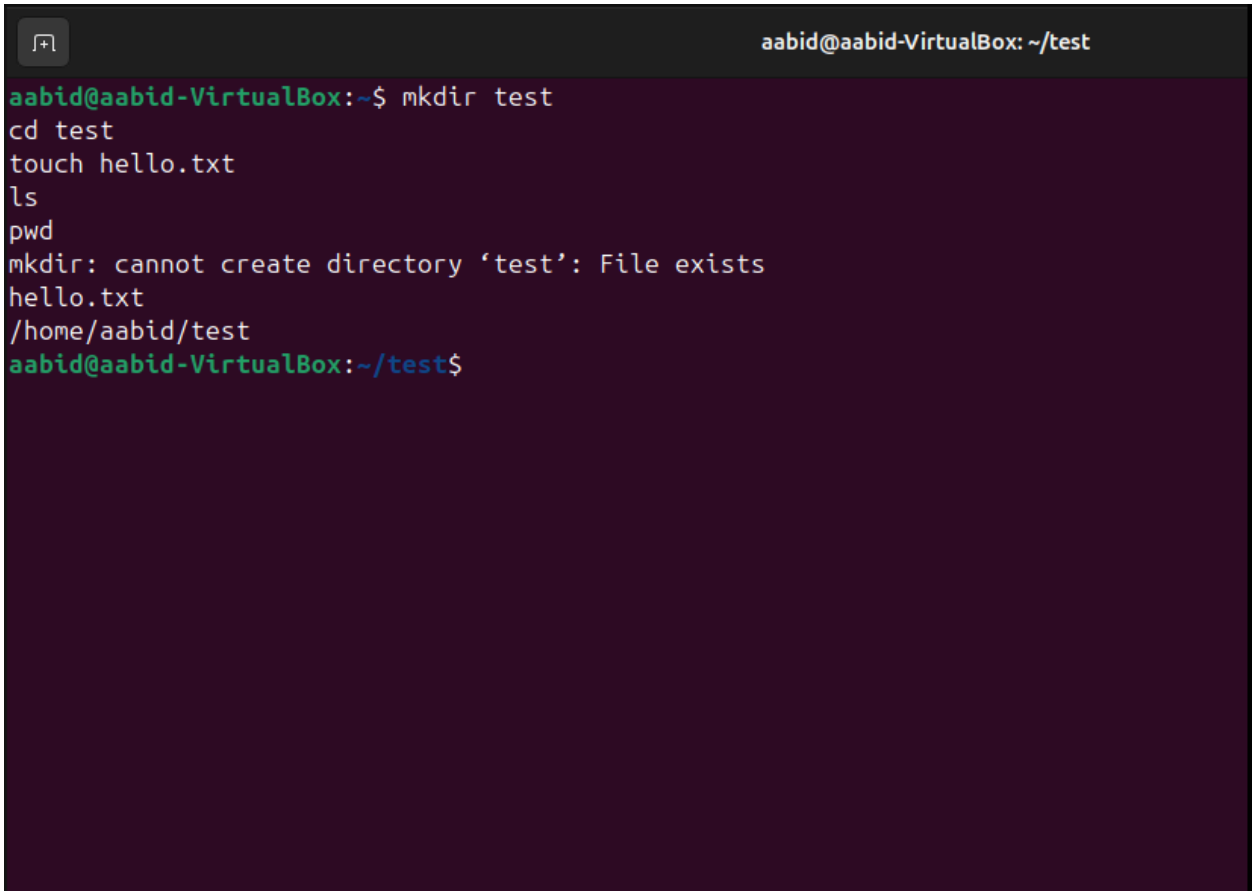
Commands used:

```
mkdir test
```

```
cd test

touch hello.txt

ls

pwd
```

These commands helped in creating directories, navigating between folders and managing files.

Figure 4: Basic Linux commands execution



3. File Permissions

Linux provides permission control for files and directories.This helps in managing security and controlling access in the system.

This command changes the permission of the file and allows it to be executed.Initially,the file did not have execution permission,but after using chmod,it was updated successfully,so now it can run.

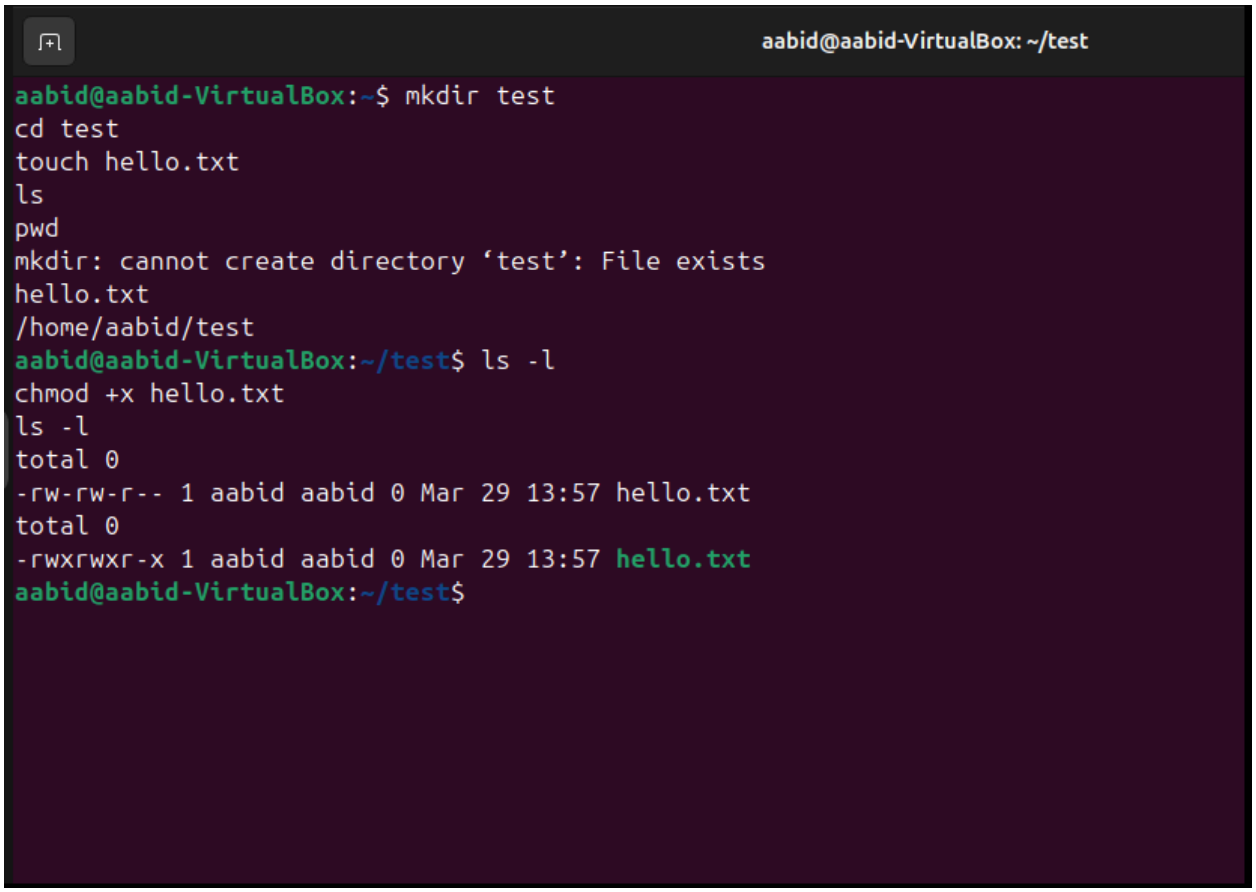
File permissions are very important in Linux as they help in controlling access and improving system security.It prevents unauthorized users from modifying important files,and this is necessary in many cases.

Permissions were checked and modified using the following **commands:**

```
ls -l

chmod +x hello.txt

ls -l
```

This shows how execution permission can be given to a file

Figure 5: File permission changes using chmod



4. User and Root Access

Linux has different types of users.Normal users have limited access while root user has full control.This process shows how a normal user can temporarily gain administrative privileges using sudo.This is useful for performing system level operations.

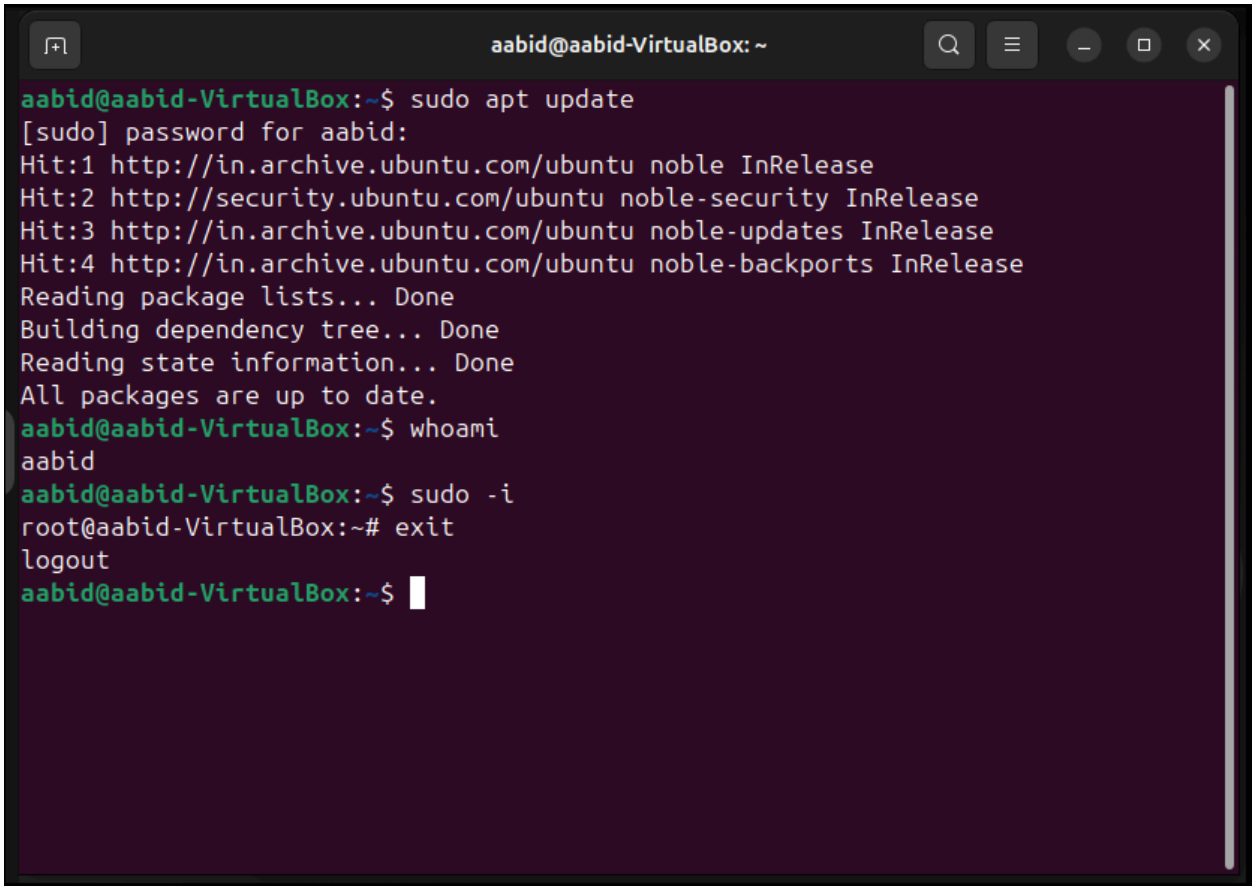
Switching back to normal user ensures that unnecessary risks are avoided,as root access can affect the entire system if misused.

Commands used:

```
whoami

sudo -i

whoami

exit

whoami
```

This shows switching between normal user and root user

Figure 6: Switching between user and root



Part C: Ecosystem,Dependencies and Community of VLC Media Player

VLC Media Player is not just a simple media player,it is actually part of a bigger open source ecosystem.What i understood while doing this is,it does not work alone,it depends on many libraries and components to run properly.

These dependencies help VLC to support many formats and features,and most of the time it works fine,but sometimes small issues can also happen,so it is not always perfect.

The ecosystem of VLC includes developers,testers and normal users from different parts of the world.All these people contribute in different ways.Some fix bugs,some add new features and some just report issues.

This makes the software better over time,but still problems can come sometimes and later they get fixed.

One important thing in VLC is FFmpeg.It is used for audio and video processing.VLC uses it to decode and encode files,so it can play many formats.

If this was not there,it would be difficult to support so many file types,and maybe users would need extra software also,and that is not convenient.

Another thing i came to know is libVLC.This is a library which allows other applications to use VLC features.So VLC is not just a player,its parts are also used in other software,and i felt this was something new for me while learning this part.

Dependencies are very important in any software.If one dependency fails,it can affect the whole system.So developers need to manage and update them properly,but still sometimes compatibility issues may happen,and that is kind of normal in real situations.

VLC also supports plugins and extensions.These help in adding extra features without changing the main software.For example subtitles or streaming options can be added,and this makes it more flexible and useful for users.

The whole VLC project is supported by an organization called VideoLAN.They manage development and distribution of VLC.They make sure everything is going properly,but still keep it open source and free,and this balance is important i think.

The development of VLC follows open source style.Anyone can contribute by giving code,reporting bugs or suggesting ideas.This makes the process open and transparent,but not all contributions are accepted directly,some checking is there before adding.

VLC also uses version control systems.This helps developers track changes and fix issues.If something goes wrong,they can go back to previous version also,and this is very useful in real development situations.

Another important thing is compatibility.VLC works on Linux,Windows and Mac without much changes.This cross platform support increases usability,and i feel this is one big reason why many people use VLC.

Community plays a very big role here.It is not controlled by a single company,but by many contributors working together.This is quite different from normal software,and gives more freedom also.

Developers from different countries contribute to VLC.They fix bugs,improve code and sometimes suggest features.This makes VLC keep improving,but development speed may not always be same,and that is ok.

Users also contribute by reporting issues and giving feedback.This helps developers understand real problems.In some cases even small feedback becomes useful,and i noticed this is important.

There are forums and discussion platforms where users and developers interact.These help in solving problems and sharing ideas,but sometimes replies may be slow,so patience is needed.

Open source community also gives transparency.Anyone can see what changes are happening and how the software is improving.

This builds trust,and not all software give this kind of openness.

Another thing is updates.VLC releases updates regularly to fix bugs and improve performance.These updates happen because of active community,but users also need to update,otherwise issues may remain.

The collaborative nature of open source projects like VLC makes them strong.Many people working together can give better result than a single team,but coordination is also needed,otherwise confusion can happen sometimes.

Documentation is also part of ecosystem.There are guides and tutorials available.

Also one thing i noticed while doing this part is,open source projects like VLC are not just about coding,it is more about how people work together.Many times different people are working on different parts,so sometimes there can be confusion or delay,but still it somehow works and improves slowly.This kind of working is little different from normal company software,and i felt it is interesting but also little bit complex to understand fully.

This helps beginners understand the software,but honestly not everyone reads documentation properly,and i also skipped sometimes while doing this.

Developers also use documentation to understand internal structure.This makes contribution easier and reduces confusion while coding,and saves time also.

Another advantage is flexibility.Users can use VLC in their own way.They are not restricted,and this becomes useful in many situations.

Another thing is dependencies management,it looks simple at first but actually it is not.If one small library is not working properly,it can affect whole software and user may not even know the reason.So developers need to check many things again and again,and sometimes issues come from unexpected places,so debugging becomes little difficult also.

VLC supports multiple platforms like Linux,Windows and Mac.This wide support is possible because of strong ecosystem and dependencies,and this increases usability overall.

Security is also important.Developers fix vulnerabilities and release updates regularly to keep software safe,but users also need to update system,otherwise problems may come.

Testing is done before updates are released.This is to make sure new changes do not break existing features.This improves reliability,but still sometimes small bugs can remain,and that is normal.

VLC also supports many languages.This is because contributors translate it.This makes it usable worldwide,and helps users from different regions,and i think this is really helpful.In my opinion,the ecosystem and community are the main reasons behind VLC success.

It is not just a software,it is a combined work of many people,and that makes it strong and reliable in long term.

This part was little long compared to others,but it helped me understand how real open source software works behind the scenes,and how many things are connected,and not everything is simple actually.

In my view,this whole ecosystem is useful but not very simple for beginners.At first i was not understanding how everything is connected,but after going through slowly it became little clear.Still i think there are many things which i may not understood fully,but overall idea is clear that open source software depends on many people and many components,and all these together make it work properly.

Part D: Comparison Between Open Source and Proprietary Software

Open source software and proprietary software are two different approaches in software development.They differ in terms of access,control and usage.

Comparison Table

DIMENSION	OPEN SOURCE SOFTWARE	PROPRIETARY SOFTWARE
Cost	Usually free	Often paid
Security (who can audit the code?)	Anyone can audit the code	Only company can audit
Support and reliability	Community support,may vary sometimes	Officials company support, more stable in many cases
Freedom to modify	Can be modified freely	Cannot be modified
Transparency	High, source code is visible	Low,source code is hidden
Vendor lock-in	Low, users can switch easily	High,users can depend on company
Learning opportunity	High,users can learn from code	Limited learning
Community vs corporate control	Controlled by community	Controlled by company
Overall verdict	More flexible and useful for learning	More controlled and stable

Open source software gives more freedom and flexibility,which is useful especially for students and developers.In many cases,it allows better understanding of how software works.

Proprietary software on the other hand provides better support and sometimes more stable experience,but it limits user control and customization.

In my opinion,open source is better for learning and development,but proprietary software may be preferred in some professional environments.

Part E: Shell Scripts

Script 1: System Identity Script

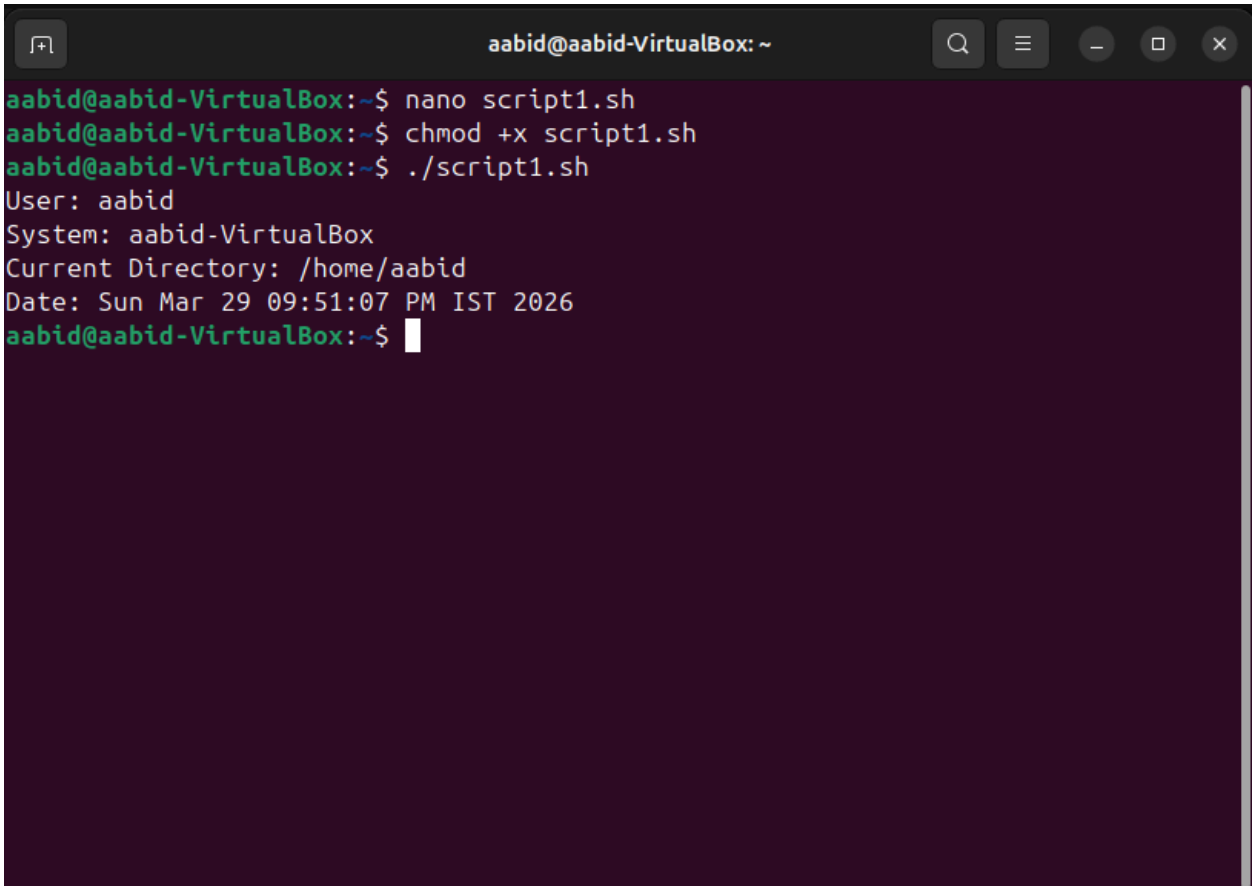
CODE:

```
#!/bin/bash
echo "User: $(whoami)"
echo "System: $(hostname)"
echo "Current Directory: $(pwd)"
echo "Date: $(date)"
```

Script 1 was created to display basic system details like user name,system name,current directory and date.This helps to understand how simple shell scripting works.

The script was executed successfully and it showed all required outputs.This part was simple and did not create any issue while running.

Figure: Script 1 showing system details output



While doing this part,I understood how multiple commands can be combined inside a script and executed together.This is useful in many cases.

At first I was not sure how it will display,but after running it became clear.

Script 2: FOSS Package Inspector

CODE:

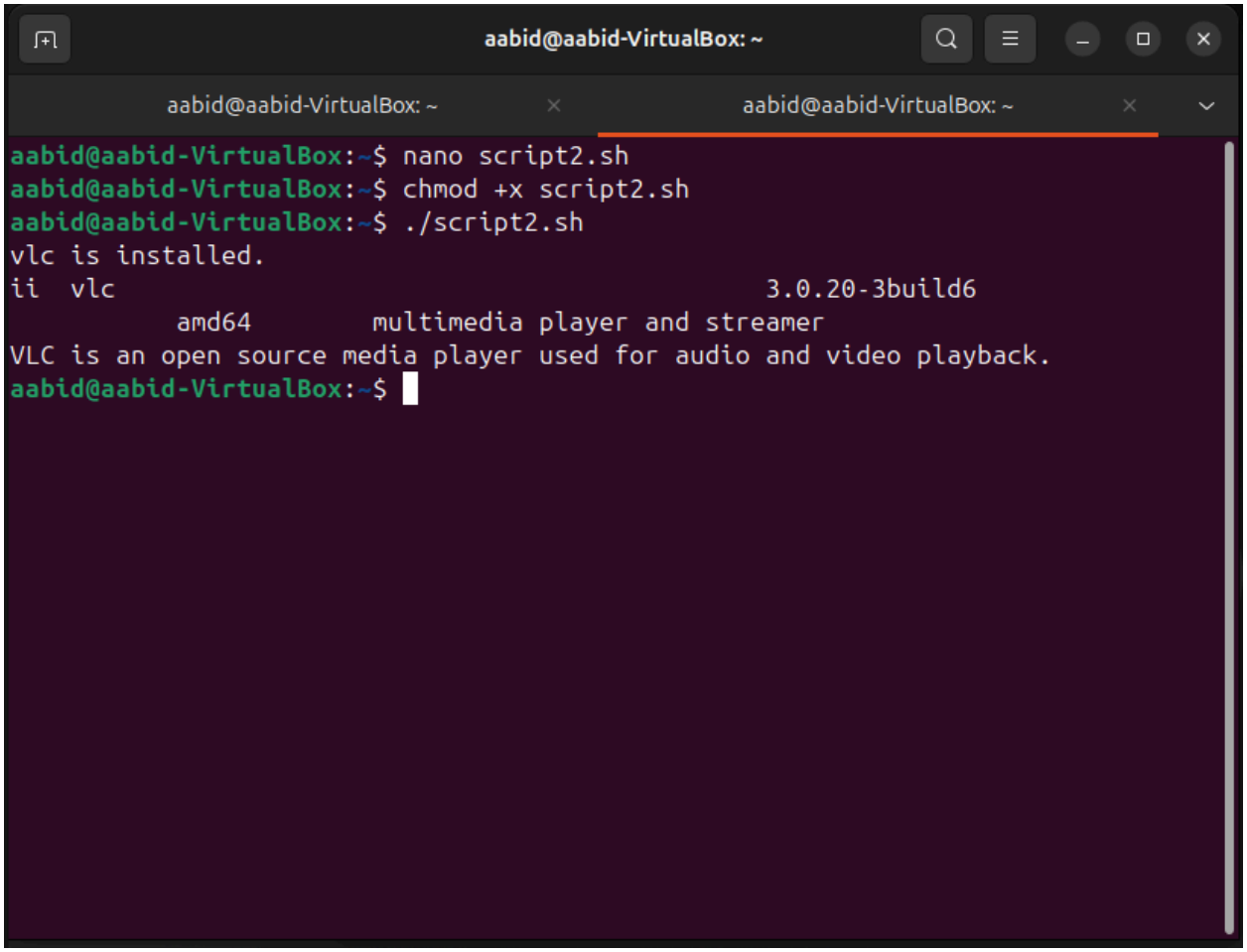
```
#!/bin/bash
PACKAGE="vlc"
if dpkg -l | grep -q "^ii  $PACKAGE "; then
    echo "$PACKAGE is installed."
    dpkg -l | grep "^ii  $PACKAGE "
else
    echo "$PACKAGE is not installed."
```

```
fi
case $PACKAGE in
    vlc)
        echo "VLC is an open source media player used for audio
and video playback."
        ;;
    firefox)
        echo "Firefox is an open source web browser."
        ;;
    git)
        echo "Git is an open source version control system."
        ;;
    *)
        echo "Unknown package."
        ;;
esac
```

Script 2 was created to check whether VLC package is installed in the system or not.It also shows package details and prints a small description using case statement.

The script worked correctly and displayed that VLC is installed along with its details.This part was little bit more interesting compared to previous script.

Figure: Script 2 showing VLC package details



While doing this,I understood how if condition and case statement works together in shell scripting,and how packages can be checked using terminal.

This part was little bit confusing at first,but after running it became clear.

Script 3: File Creator using Loop

CODE:

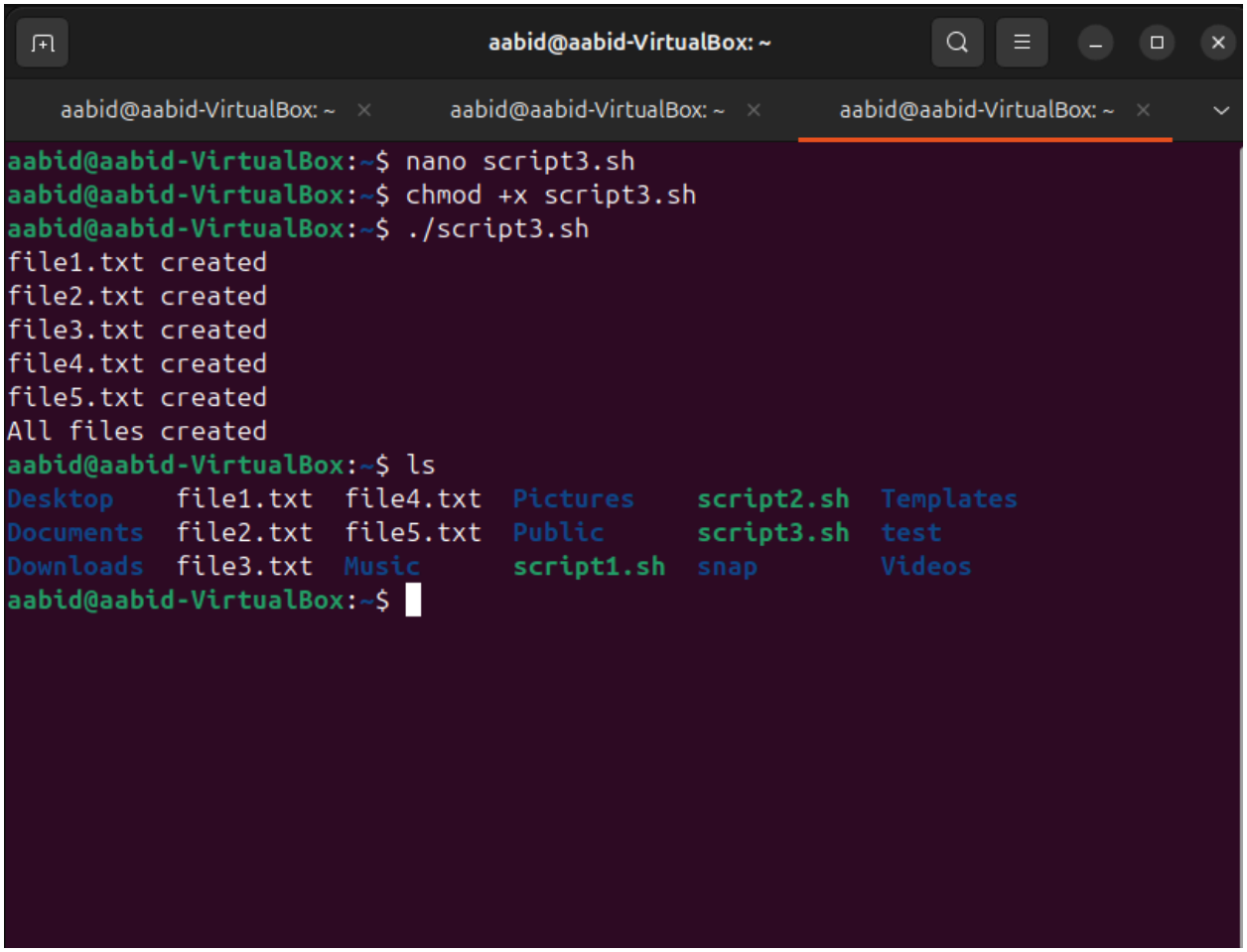
```
#!/bin/bash
for i in 1 2 3 4 5
do
    touch file$i.txt
```

```
    echo "file$i.txt created"
done
echo "All files created"
```

Script 3 was created to generate multiple files using loop.This helps to understand how repetition works in shell scripting.

The script created files automatically and displayed messages for each file.This makes the task easier instead of creating files one by one,and it saves time also.

Figure: Script 3 showing file creation using loop



While doing this script,I understood how loops work in shell scripting.This part was simple but still needed attention while typing loop.

At first I thought files will overwrite,but later I understood it creates new ones.

Script 4: Even or Odd Checker

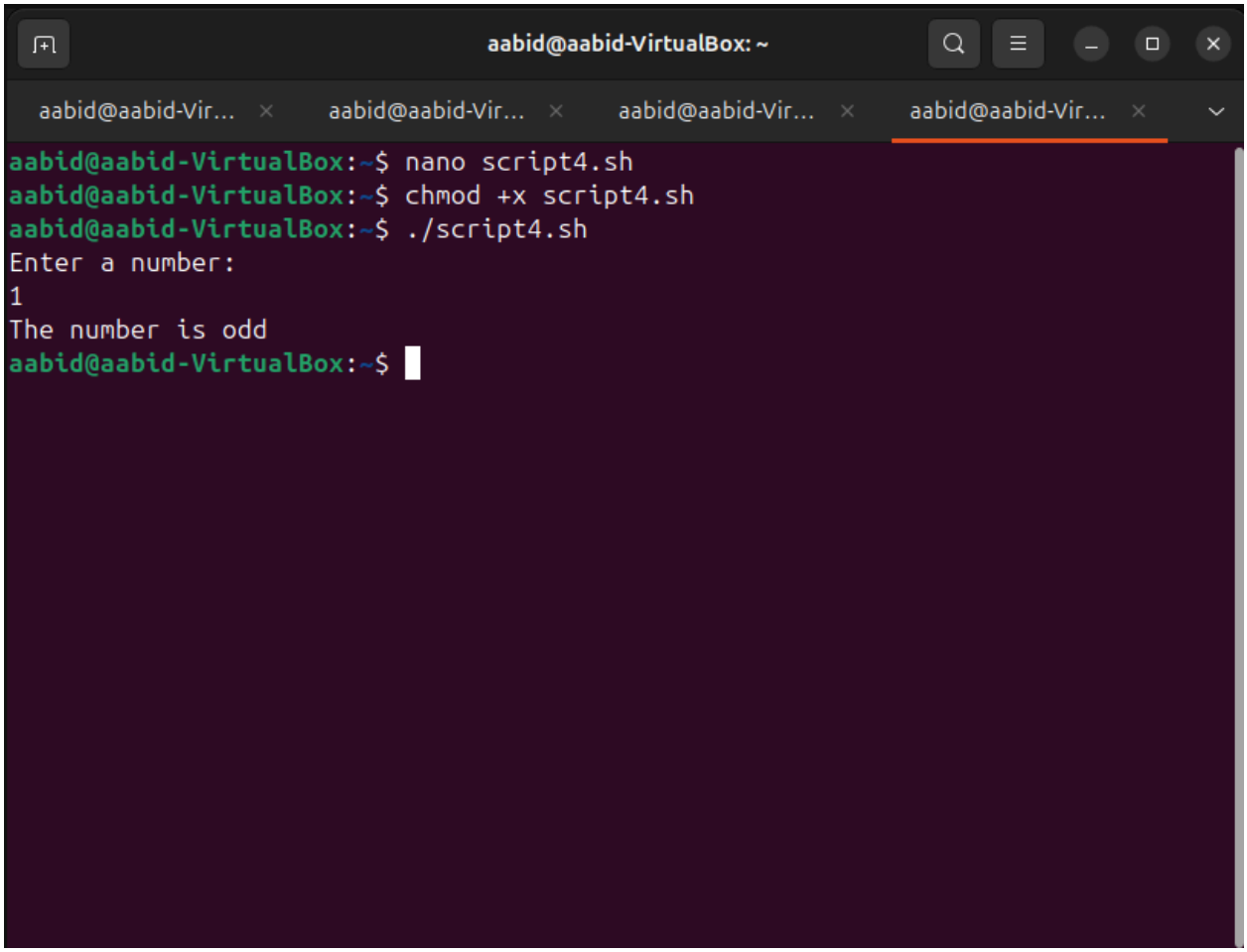
CODE:

```
#!/bin/bash
echo "Enter a number:"
read num
if [ $((num % 2)) -eq 0 ]
then
    echo "The number is even"
else
    echo "The number is odd"
fi
```

Script 4 was created to check whether a number is even or odd by taking input from user.This helps in understanding how input and conditions work in shell scripting.

The script worked correctly and displayed result based on the number entered.This part was slightly different because it required user input.

Figure: Script 4 showing even or odd result



While doing this,I understood how read command works and how if condition is applied in shell scripting.It was not difficult but needed careful typing.

At first I tried with wrong input,but later it worked properly.

Script 5: Simple Calculator

CODE:

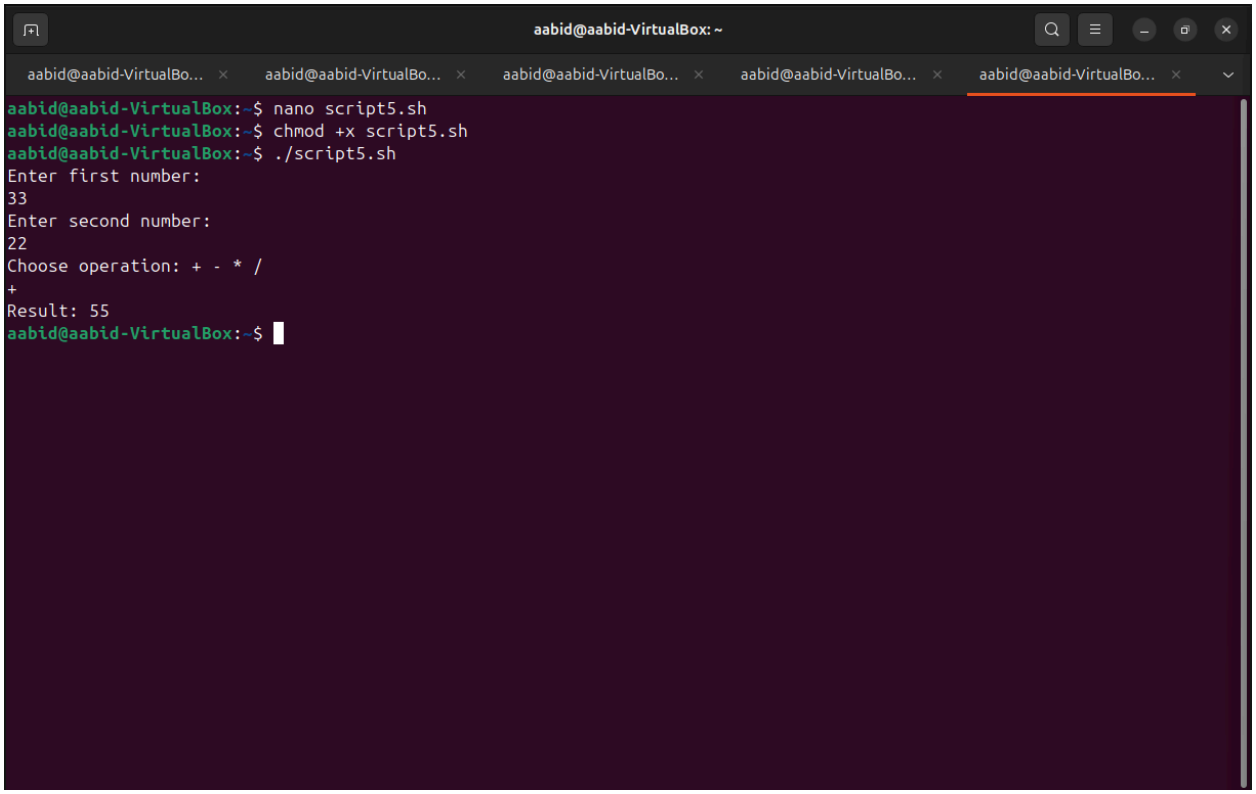
```
#!/bin/bash
echo "Enter first number:"
read num1
```

```
echo "Enter second number:"
read num2
echo "Choose operation: + - * /"
read op
case $op in
    +)
        echo "Result: $((num1 + num2))"
        ;;
    -)
        echo "Result: $((num1 - num2))"
        ;;
    \*)
        echo "Result: $((num1 * num2))"
        ;;
    /)
        echo "Result: $((num1 / num2))"
        ;;
    *)
        echo "Invalid operation"
        ;;
esac
```

Script 5 was created to perform basic arithmetic operations using user input.This includes addition,subtraction,multiplication and division.

The script worked correctly and displayed result based on selected operation.This part was more interesting because it combined input,case statement and calculations together.

Figure: Script 5 showing calculator result



While doing this,I understood how different operations can be handled using case statement and user input.This part was little bit confusing while typing symbols,but later it worked.

At first I entered wrong symbol,but then corrected it and got output.

Final Understanding

Through these scripts,I understood how basic shell scripting works in Linux.It helped me to learn commands,conditions,loops and user input handling.

This section was overall interesting and practical compared to theory parts,and it gave better understanding of how things actually work in system.

Overall,this project helped me to understand open source software in both practical and theoretical way,and it improved my confidence in using Linux system.